

Amazon Product Sales Data Analytics -- Mini Project Report

Project Title

Amazon Product Sales Data Analytics

Project Objective

The objective of this mini project is to analyze Amazon product sales and performance data using **Microsoft Excel** and **Power BI**.

The project focuses on cleaning and preprocessing the raw dataset in Excel, handling missing values and duplicates, standardizing data, creating calculated fields, and then using the cleaned data in Power BI to build interactive dashboards.

The analysis covers **products, pricing, ratings, discounts, popularity, promotions, and availability**.

Dataset Description

Particular Details

Dataset Name Amazon Product Sales Data 42K+ Items -- 2025

Domain Sales & E-commerce

Source Kaggle

Number of Records 42,675

Number of Columns 16

Dataset Type Raw / Uncleaned

Tools Used Excel & Power BI

Dataset Source:

<https://www.kaggle.com/datasets/srisyra02/amazon-product-sales-data-42k-items-2025>

The dataset contains Amazon product information related to ratings, customer reviews, purchases, pricing, discounts, promotional activity, best-seller status, sponsorship, coupon availability, Buy Box availability, delivery information, and other product attributes.

Methodology

The project was completed in two major phases:

Phase 1 -- Excel Data Cleaning & Preprocessing

The raw Amazon dataset was cleaned and standardized using Microsoft Excel.

Phase 2 -- Power BI Analysis & Dashboard Development

The cleaned Excel data was imported into Power BI, where calculated columns, DAX measures, interactive visualizations, slicers, and drill-through functionality were created.

Excel Data Cleaning & Preprocessing

5.1 Duplicate Check

The dataset was checked for duplicate records using Excel's **Remove Duplicates** feature.

All 16 original columns were selected for the duplicate check. No duplicate records were found, so no rows were removed.

5.2 Missing Value Analysis

Missing values were identified using the **COUNTBLANK** function.

Important numerical fields such as Rating, Number of Reviews, Monthly Purchase, Discounted Price, and Original Price were checked and handled based on the available information.

5.3 Rating and Number of Reviews

Missing values in **Rating** and **Number of Reviews** were handled using **median imputation** because these fields can contain extreme values.

- Rating median = **4.5**
- Number of Reviews median = **343**

5.4 Monthly Purchase Conversion

The `bought_in_last_month` field contained text values such as:

- 300+ bought in past month
- 10K+ bought in past month

These values were converted into numerical values in a new **Monthly Purchase** column.

Where purchase information was unavailable, **0** was used and a separate **Purchase Data Status** field was created to distinguish available and unavailable purchase information.

5.5 Price Cleaning

The `price_on_variant` field contained a mixture of text and price information.

The relevant price information was extracted and converted into numerical format. Missing discounted prices were filled using the available variant price where appropriate.

After the required price information had been captured in the final price fields, the `price_on_variant` column was removed because it was not required for the final analysis.

5.6 Original Price Cleaning

Currency symbols and commas were removed from the **Original Price** field to convert it into numerical format.

Where **No Discount** was present, the applicable selling price was used as the Original Price.

5.7 Delivery Date Cleaning

The original **Delivery Date** field contained several **Not Available** values.

Instead of creating estimated delivery dates, the available **collected_at** date was used to fill missing delivery information. A new **Cleaned Delivery Date** column was created while retaining the original Delivery Date column for reference.

5.8 Buy Box Availability

The **Buy Box Availability** field contained many **Not Available** values.

These were retained as **Not Available** because they represent unavailable information and should not be treated as a numerical zero.

5.9 Excel Pivot Tables and Charts

Pivot Tables and Pivot Charts were created in Excel to perform initial analysis.

The main analyses included:

- **Product Count by Rating** -- shows the number of products at each rating level.
 - **Sponsored vs Organic Products** -- compares sponsored and organic product listings.
 - **Rating vs Average Number of Reviews** -- compares ratings with average customer review volume to understand the relationship between ratings and popularity.
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Power BI Data Preparation

The cleaned Excel dataset was imported into Power BI.

Unnecessary blank columns were removed or hidden, and the required fields were checked to ensure appropriate data types.

6.1 Calculated Columns

The following calculated columns were created:

Discount %

Calculates the percentage discount for each product using its Original Price and Discounted Price.

Price Category

Products were grouped into:

- Low
- Medium
- High

based on discounted price.

Rating Category

Products were grouped into:

- Poor
- Average
- Good
- Excellent

based on their rating.

Review Category

Products were grouped into:

- Low
- Moderate
- High
- Very High

based on the number of customer reviews.

Power BI Measures

DAX measures were created to support KPI cards and dashboard analysis.

Product Measures

- Total Products
- Average Rating
- Total Reviews
- Average Reviews

Purchase Measures

- Total Monthly Purchases
- Average Monthly Purchases

Pricing Measures

- Average Original Price
- Average Discounted Price
- Average Discount %
- Maximum Discount %

Performance Measures

- Best Seller Products
- Sponsored Products
- Couponed Products

These measures allow the dashboard values to update dynamically when users interact with slicers.

Power BI Dashboard Description

The Power BI report contains **six pages** designed to provide different levels of analysis.

8.1 Page 1 -- Executive Overview

Purpose

The Executive Overview provides a high-level summary of the Amazon product dataset and allows users to quickly understand overall product performance.

KPI Cards

Five KPI cards are used:

1. **Total Products**
Displays the total number of products in the dataset.

2. **Average Rating**
Shows the overall average customer rating.
3. **Total Reviews**
Shows the total number of customer reviews across the products.
4. **Average Monthly Purchases**
Shows the average number of monthly purchases per product based on the prepared purchase data.
5. **Average Discount %**
Shows the average percentage discount across products.

Visualizations

Product Count by Rating

Shows how products are distributed across individual rating values and helps identify the most common rating levels.

Sponsored vs Organic Products

Compares the number of organic and sponsored product listings.

Couponed vs Non-Couponed Products

Shows how many products have coupons compared with products without coupons.

Products by Price Category

Displays the distribution of products across Low, Medium, and High price categories.

Slicers

- Rating
- Sponsored Status
- Coupon Status
- Best Seller Status

These slicers allow users to filter the entire page and explore specific groups of products.

Navigation

A page navigator allows users to move between the major analytical pages of the report.

Page 2 -- Product Analytics

Purpose

The Product Analytics page provides a more detailed analysis of product ratings, customer reviews, and purchasing behavior.

KPI Cards

1. **Average Rating** -- overall average product rating.
2. **Total Reviews** -- total customer review volume.
3. **Average Reviews per Product** -- average number of reviews received per product.
4. **Average Monthly Purchases** -- average monthly purchase volume.

Visualizations

Product Count by Rating Category

Shows the number of products classified as Poor, Average, Good, and Excellent.

Average Reviews by Rating Category

Compares the average number of reviews received by each rating category.

Rating vs Number of Reviews

A scatter chart used to explore the relationship between product ratings and customer review volume.

Average Monthly Purchases by Rating Category

Compares average monthly purchases across Poor, Average, Good, and Excellent rating categories.

Slicers

- Rating Category
- Price Category
- Sponsored Status

These filters help users examine customer engagement and purchasing behavior for selected product groups.

Page 3 -- Pricing & Discount Analytics

Purpose

This page focuses on product pricing, discounts, and promotional activity.

KPI Cards

1. **Average Original Price** -- average original/listed price.
2. **Average Discounted Price** -- average selling price after discount.
3. **Average Discount %** -- average percentage discount.
4. **Maximum Discount %** -- highest discount percentage available in the dataset.

Visualizations

Average Price by Rating

Compares average product price across rating levels.

Average Discount % by Rating

Shows how discount percentages vary across product ratings.

Price vs Number of Reviews

A scatter chart used to examine the relationship between product price and customer review volume.

Couponed vs Non-Couponed Products

Compares products with coupons against products without coupons.

Slicers

- Rating
- Coupon Status
- Sponsored Status
- Price Category

These allow users to analyze pricing and promotions for specific product groups.

Page 4 -- Performance Analysis

Purpose

The Performance Analysis page evaluates product demand, customer engagement, best-seller performance, and product characteristics.

KPI Cards

1. **Total Products**
2. **Average Monthly Purchases**
3. **Average Rating**
4. **Total Reviews**

5. Best Seller Products

These KPIs provide a quick overview of product performance.

Visualizations

Average Monthly Purchases by Rating Category

Compares product demand across rating categories.

Sponsored vs Organic Products

Shows the distribution of sponsored and organic listings.

Total Reviews by Rating Category

Compares total customer review volume across rating categories.

Products by Price Category

Shows the number of products in Low, Medium, and High price categories.

Best Seller vs Other Products

Compares products identified as Best Sellers with other products.

Rating vs Average Monthly Purchases

Examines the relationship between customer ratings and average monthly purchase activity.

Slicers

- Rating Category
 - Price Category
 - Sponsored Status
 - Coupon Status
 - Best Seller Status
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Page 5 -- Product Details

Purpose

The Product Details page provides detailed information about an individual product.

This page uses **Power BI drill-through functionality**.

KPI Cards

1. **Product Rating**

2. **Total Reviews**
3. **Monthly Purchases**
4. **Discounted Price**

These cards display metrics for the selected product.

Product Information

The detailed product information section displays:

- Product Title
- Rating
- Number of Reviews
- Monthly Purchases
- Original Price
- Discounted Price
- Discount %

Product Status

The page also displays the selected product's:

- Coupon Status
- Sponsored Status
- Best Seller Status

Drill-Through Functionality

The **Product Title** field is used as the drill-through field.

Users can right-click a product in an applicable visual and select:

Drill-through → Product Details

The selected product title is passed as a filter, allowing the page to dynamically display the selected product's rating, reviews, purchases, pricing, and promotional status.

A back button allows users to return to the previous analysis page.

Page 6 -- Insights & Recommendations

Purpose

The final page summarizes the major findings from the analysis and provides actionable recommendations.

Key Insights

Strong Product Ratings

The average product rating is **4.40**, with most products falling into the **Excellent** rating category.

High Review Volume

The dataset contains approximately **129 million total reviews**, indicating substantial customer engagement across the products.

Limited Coupon Usage

Approximately **98% of products have no coupon**, indicating that coupon-based promotions are used by a relatively small share of products.

Mostly Organic Listings

Around **96% of products are classified as Organic**, while approximately **4% are Sponsored**.

Recommendations

Strengthen High-Rated Products

Maintain product quality and increase visibility for products with strong customer ratings.

Optimize Discount Strategy

Analyze products with higher discounts and monthly purchases to identify discount levels that may help improve demand.

Improve Low-Rated Products

Identify lower-rated products and focus on product quality, product descriptions, and the overall customer experience.

Leverage Customer Reviews

Use products with strong review volumes and ratings as benchmarks to understand factors associated with better product performance.

Key Findings

The analysis indicates that the Amazon product dataset contains a strong concentration of highly rated products and a substantial volume of customer reviews.

The analysis also shows that most listings are organic rather than sponsored and that coupon usage is limited.

Pricing and discount analysis provides a way to compare product prices and promotional activity with ratings and review volume, while purchase-related analysis helps evaluate product demand.

The dashboard enables users to investigate these patterns interactively using slicers and product-level drill-through.

Recommendations

Based on the analysis:

1. **Maintain high product quality** to support strong ratings and customer satisfaction.
 2. **Increase visibility of highly rated products** through appropriate promotional and marketing strategies.
 3. **Optimize discount levels** by comparing discounts with purchase activity rather than applying large discounts uniformly.
 4. **Investigate lower-rated products** and improve product quality, descriptions, and customer experience.
 5. **Use customer reviews as a source of product feedback** and identify characteristics associated with strong-performing products.
 6. **Evaluate sponsored promotions carefully** and compare their performance with organic listings.
 7. **Consider targeted coupon strategies** for products where promotional incentives may help increase demand.
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Conclusion

This mini project demonstrates an end-to-end data analytics workflow using **Excel and Power BI**.

The raw Amazon product dataset was first checked for duplicates and missing values and then cleaned and standardized in Excel. Important fields such as ratings, reviews, monthly purchases, prices, and delivery dates were prepared for analysis.

The cleaned data was then imported into Power BI, where calculated columns, DAX measures, interactive visualizations, slicers, navigation, and drill-through functionality were implemented.

The final six-page dashboard provides a comprehensive view of Amazon product performance, covering overall performance, product analytics, pricing and discounts, demand and engagement, individual product details, and key insights with recommendations.

The project demonstrates how data cleaning, analysis, visualization, and business-oriented insight generation can be combined to turn raw e-commerce data into an interactive analytical report.